Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, T_{2u} doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per `Docs/status/TOE-FACT-SHEET.md` should be re-calibrated against Math411-AddB §10.

Pillar 5 of the Topological Energy Condensate Theory: chirality from shell-protected Dirac genericity, opposite-valley pairing, mass protection, and the reduced Weyl/Dirac witness theorem on the BCC condensate

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We present the Pillar-5 chirality programme of the Topological Energy Condensate Theory (TECT). The candidate mechanism combines five layered results from the Math10–14 + Math171-AddA canonical archive: (i) Math10 shell-genericity — once the Bragg mass is tuned away, emergent Dirac propagation is generic in the symmetry-allowed tensor space (Single-Channel and Two-Channel Genericity Theorems + Dirac Genericity Principle); (ii) Math11 minimal-model uniqueness — five microscopic constraints (finite-q shell instability, quartic stabilisation, internal $\mathbb{CP}^2$ geometry, enriched shell-Dirac mechanism, second-order time dynamics) force the minimal field content to be the complex triplet; (iii) Math12 mass protection — on the doubled block (after opposite-valley pairing) the classification of constant Hermitian operators identifies a four-dimensional symmetry-preserving space, of which only two elements are genuine Dirac masses; the residual valley charge generator forbids the remaining channels; (iv) Math13 local Dirac block reduction — under the residual little-group mismatch, the local Dirac carrier reduces to $\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}$, reducing the first remaining Dirac-closure condition to a finite microscopic audit; (v) Math14 reduced Weyl/Dirac witness theorem — three theorems (linearised quartic no-go, exact shell cancellation of the isotropic first-order term at $q_*^2 = -Z/(2Y)$, locked cross-shell quartic vanishing at zeroth order) close the reduced Weyl/Dirac witness inside the Class II seed space; (vi) Math171-AddA Hirzebruch–Riemann–Roch index formula (Top-impact paper TI-1): $\mathrm{ind}(D_E^c) = r - \mu$ on $\mathbb{CP}^2$, specialising to ind $= 16$ at the rank-16 Standard-Model fermion sector with $\mu = 0$ (NOT the previously incorrect $\mathrm{Index}(D) = \mathrm{wind}(m)$ identification on the 3D Brillouin torus, which would have required an unavailable Callias / domain-wall framework). The combined statement is a Pillar-5 *candidate mechanism note* at the canonical `Docs/status/TOE-FACT-SHEET.md` tier, NOT a closed theorem of the form "the chiral family count equals the winding number". A theorem-level rewrite within a Callias / Niemi–Semenoff / domain-wall framework on the appropriate non-compact extension remains an open programme item per `Q-2026-05-02-Pillar5-Mechanism-Rewrite`.

I. INTRODUCTION AND THE STRUCTURAL QUESTION

Chirality — the asymmetric coupling of left- versus right-handed fermions to weak-scale gauge interactions — is one of the most striking features of the Standard Model. The TECT framework addresses the candidate *origin* of chirality: rather than postulating left-handed Weyl spinors as fundamental input fields, the candidate proposal is that chirality emerges from the topological-defect sector of the BCC condensate via shell-protected Dirac couplings, opposite-valley pairing, and a residual-valley-symmetry mass protection.

The present paper reports the Pillar-5 chirality programme as a *candidate mechanism note* (Rev 3, 2026-05-03) compiling the canonical Math10–14 + Math171-AddA content. We do NOT claim a closed theorem of the form "$\mathrm{Index}(D) = \mathrm{wind}(m)$" on the 3D Brillouin torus: such an identification would require a Callias / Niemi–Semenoff / domain-wall framework that is not provided by the present archive and is tracked separately as an open follow-up (`Q-2026-05-02-Pillar5-Mechanism-Rewrite`).

The structural question is therefore: *what features of the BCC condensate machinery are necessary, what is sufficient, and what remains conjectural*? The paper organises the five Math10–14 results into a layered candidate mechanism (§§II–VI), relates the rank-16 family-count specialisation to the corrected HRR formula of Math171-AddA / Top-impact paper TI-1 (§VII), and states the honest scope explicitly in §VIII.

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II. MATH10 SHELL-GENERICITY: GENERIC NONVANISHING OF SHELL-PROTECTED DIRAC COUPLINGS

The first layer establishes that, once the Bragg mass at the locked shell is tuned away, the emergence of Dirac-type linear dispersion in the shell-protected band is generic in the symmetry-allowed microscopic tensor space.

Theorem 1 (Single-channel Dirac genericity, after Math10). *Let $\mathcal{T}_{\mathrm{sym}}$ be the linear space of microscopic gradient-linear tensor couplings of the form N^a_{ij}, where i is the output spatial index, j the gradient index, and a the internal channel label, restricted to the O_h-equivariant subspace allowed by the Brazovskii operating-point symmetries. Then the locus inside $\mathcal{T}_{\mathrm{sym}}$ on which the shell-protected Dirac velocity in any single channel vanishes is of strictly positive codimension; consequently the velocity is* generic *(i.e., non-vanishing on a Zariski-dense open subset).*

Theorem 2 (Simultaneous two-channel genericity, after Math10). *Under the same symmetry conditions, the simultaneous vanishing of the Dirac velocities in two distinct shell-protected channels is of strictly positive codimension in $\mathcal{T}_{\mathrm{sym}}$; consequently the two-channel Dirac structure is generic.*

Principle 3 (Dirac Genericity Principle, after Math10). Once the Bragg mass is tuned away and the microscopic enriched tensors are not symmetry-forbidden, emergent Dirac propagation is generic in the symmetry-allowed tensor space.

Remark 4 (Status of the genericity theorems). The Math10 theorems are *transversality / dimension-count* results in the space of allowed microscopic couplings; they do NOT by themselves construct the Standard-Model gauge structure or the chiral family count. They establish that the Dirac mechanism is not forbidden by any obvious symmetry obstruction.

III. MATH11 MINIMAL-MODEL UNIQUENESS: THE COMPLEX TRIPLET

The second layer addresses the question: *which microscopic field content is forced* by the conjunction of the Brazovskii shell-instability mechanism, quartic stabilisation, internal geometric structure, and the Math10 shell-Dirac mechanism?

Theorem 5 (Math11 minimal-model uniqueness; informal statement). *Among candidate microscopic models compatible with the conjunction of:*

1. *Constraint I — finite-q shell instability with locked shell radius q_0 from Brazovskii kinetics;*

2. *Constraint II — quartic stabilisation with $g > 0$ ensuring energy boundedness from below;*

3. *Constraint III — internal $\mathbb{CP}^2$ geometric structure forced by the locked-triple algebra;*

4. *Constraint IV — the enriched shell-Dirac mechanism (Math10) requiring an enriched internal doublet plus gradient-linear microscopic tensor couplings;*

5. *Constraint V — second-order time dynamics consistent with relativistic kinematics at the IR scale;*

the minimal field content is the complex triplet $\Psi : \mathbb{R}^3 \to \mathbb{C}^3$. Lower-complexity alternatives (real scalar, single complex scalar, complex doublet) fail at least one of Constraints I–V, and higher-complexity alternatives (complex quartet, etc.) are non-minimal.

Remark 6 (Reading guide)*.* The Math11 theorem is a *constraint-counting* statement on the microscopic field content, given the Brazovskii + shell-Dirac + internal-$\mathbb{CP}^2$ inputs. The reader should view it as the TECT-internal answer to the question "why this particular field content?", not as an external no-go argument against general alternative theories.

IV. MATH12 MASS PROTECTION: CLASSIFICATION OF CONSTANT DIRAC MASSES ON THE DOUBLED BLOCK

After opposite-valley pairing on the locked shell, the shell-relevant local Dirac block is doubled into a $\mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}$ structure. Math12 classifies the constant Hermitian operators on this doubled block compatible with the residual symmetry.

Lemma 7 (Classification of constant Dirac masses, after Math12)**.** *Let K be a constant Hermitian operator on the doubled block $\mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}$ that preserves the little-group action at the band-crossing point. By Schur's lemma applied to the irreducible little-group representation ρ on each valley, the space of symmetry-preserving constant operators is four-dimensional:*

$$K = \alpha\, \tau_0 \otimes \nVdash + \beta\, \tau_3 \otimes \nVdash + \gamma\, \tau_1 \otimes \nVdash + \delta\, \tau_2 \otimes \nVdash,$$

where τ_i are Pauli matrices on the valley pair $(+G_*, -G_*)$. Of these four parameters, only two (γ, δ in the above basis) are genuine Dirac-mass candidates that open a chiral gap on the doubled block; the remaining two (α, β) are energy-shift / valley-energy-splitting terms, NOT genuine Dirac masses.

Theorem 8 (Mass protection from residual valley symmetry, after Math12)**.** *The residual valley charge generator $Q_V := \tau_3 \otimes \mathbb{1}$ commutes with the shell Dirac kinetic operator and forbids the addition of the genuine Dirac-mass channels at the constant-operator level. Consequently, on the doubled block, the chiral zero modes are protected against the addition of constant mass terms by the residual valley symmetry; their gap can only be opened by symmetry-breaking interactions of higher order (non-constant, non-symmetric) that are not generic in the allowed microscopic-coupling space.*

V. MATH13 LOCAL DIRAC BLOCK REDUCTION $\mathcal{H}_D \cong \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}}$

The third layer reduces the first remaining Dirac-closure condition to a finite microscopic audit on the doubled block.

Proposition 9 (Math13 local Dirac block reduction)**.** *Under the little-group mismatch obtaining at the locked Brazovskii operating point, the actual local Dirac carrier (the shell-relevant local block $\mathcal{H}_D$) is NOT forced by an ordinary 2D little-group irreducible representation. Instead, the local Dirac carrier reduces to the tensor product*

$$\mathcal{H}_D \;\cong\; \mathbb{C}^2_{\text{valley}} \otimes \mathbb{C}^2_{\text{int}},$$

where:

- $\mathbb{C}^2_{\text{valley}}$ *is the opposite shell pair* $(+G_*, -G_*)$*;*
- $\mathbb{C}^2_{\text{int}}$ *encodes the residual grading, parity, or chirality at each valley.*

The first remaining full Dirac-closure condition is reduced to a finite microscopic audit on this four-dimensional carrier.

Remark 10 (On the absence of an Atiyah–Singer-type closed-form formula). The reduction to the finite carrier $\mathcal{H}_D$ is NOT an Atiyah–Singer-type identity of the form "Index(D) = topological invariant" on the 3D Brillouin torus. Such an identity would require a Callias / Niemi–Semenoff / domain-wall framework on a non-compact extension or a band-projector with controlled IR boundary; the present BCC framework does not by itself provide this. The Math13 proposition is therefore a *block-reduction* statement, not an index identity.

VI. MATH14 REDUCED WEYL/DIRAC WITNESS THEOREM: THREE CANCELLATION RESULTS

The fourth layer provides three explicit theorems closing the reduced Weyl/Dirac witness inside the Class II seed space at the Brazovskii-locked operating point.

Theorem 11 (Linearised quartic no-go, after Math14 Theorem 1)**.** *At linear order in the fluctuation $\delta\Psi$ around the locked Brazovskii background $\Psi_0(x) = A\, e^{iG\cdot x}\, z_0$ (with $|G| = q_*$, $q_*^2 = -Z/(2Y)$, $z_0^\dagger z_0 = 1$), the quartic self-coupling $g(\Psi^\dagger\Psi)\Psi$ does NOT directly generate any enriched first-order operator of the form $(\partial_i\Psi_0)(-i\partial_j) \otimes s^a$. Consequently the quartic does not directly produce an enriched first-order tensor N^a_{ij} at linear order.*

Theorem 12 (Exact shell cancellation of the isotropic first-order term, after Math14 Theorem 2)**.** *For the minimal isotropic kinetic operator $Z\nabla^2 - Y\nabla^4$, the shell-projected first-order transport term in the envelope χ vanishes exactly at the Brazovskii shell $q_*^2 = -Z/(2Y)$.*

Theorem 13 (Locked cross-shell quartic vanishing at zeroth order, after Math14 Theorem 3)**.** *The locked cross-shell quartic projection in the shell-restricted envelope theory vanishes at zeroth order in the small-detuning expansion. Consequently, the cross-shell quartic does NOT contribute to the enriched first-order tensor structure required for the shell-Dirac mechanism at zeroth order.*

Corollary 14 (Reduced Weyl/Dirac witness closed in Class II seed space, after Math14 Corollary)**.** *The conjunction of Theorem 11, Theorem 12, and Theorem 13 closes the reduced Weyl/Dirac witness theorem inside the explicit Class II seed space at the Brazovskii-locked operating point: the obstructions to the shell-Dirac mechanism at the relevant order are eliminated, leaving the genericity arguments of Math10 (§II) unblocked.*

VII. MATH171-ADDA HIRZEBRUCH–RIEMANN–ROCH INDEX FORMULA AND THE RANK-16 SPECIALISATION

The chiral family count is addressed not by a winding-number identification on the 3D Brillouin torus, but by the Hirzebruch–Riemann–Roch (HRR) index formula on the canonical Pillar-4 realisation base, which reduces to a rank specialisation of the general spin-c index.

Theorem 15 (Math171-AddA HRR index formula, see Top-impact paper TI-1 for full statement)**.** *For a holomorphic vector bundle E of rank r on $\mathbb{CP}^2$ with $c_1(E) = 0$ and $c_2(E) = \mu H^2$ for some* *integer $\mu \in \mathbb{Z}$ (where H is the hyperplane class), the spin-c Dirac index with the canonical spin-c structure is*

$$\mathrm{ind}(D_E^c) \;=\; r - \mu \;\in\; \mathbb{Z}.$$

Specialising to the rank-16 SO(10) chiral-fermion sector $r = 16$ at the canonical realisation $\mu = 0$ (Math179–198 archive internal reading) gives $\mathrm{ind}(D_E^c) = 16$, consistent with one Standard-Model family per chiral irrep at the level of the historical $\mathbb{CP}^2$-route algebraic count.

Remark 16 (Replacement of the Index(D) = wind(m) identification). The previously incorrect identification "Index(D) = wind(m)" on the 3D Brillouin torus, applied to a real scalar mass field $m(\mathbf{k})$ without a Callias-type / domain-wall framework, does not hold in general. The correct chiral-count machinery in the TECT framework is the conjunction of (a) the Math10 shell-genericity theorems (§II), (b) the Math12 mass protection (§IV), (c) the Math13 local Dirac block reduction $\mathcal{H}_D \cong \mathbb{C}^2_{\mathrm{valley}} \otimes \mathbb{C}^2_{\mathrm{int}}$ (§V), and (d) the HRR-formula family-count specialisation $\mathrm{ind}(D_E^c) = r - \mu$ at $r = 16$, $\mu = 0$ (Theorem 15). The historical Pillar-4 $\mathbb{CP}^2$-route framing is preserved in the companion top-impact paper TI-1; the canonical Pillar-4 realisation programme operates on $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ per Math305 (see Paper 04 for the full Pillar-4 atomic theorem).

VIII. HONEST SCOPE: WHAT PILLAR 5 ESTABLISHES AND WHAT REMAINS OPEN

a. What the present paper establishes. At the canonical mechanism-note tier per the canonical `Docs/status/TOE-FACT-SHEET.md`, the present paper compiles a five-layer candidate Pillar-5 chirality mechanism: (i) shell-Dirac genericity (Math10 Theorems 1–2 and Principle 3); (ii) minimal-model uniqueness forcing the complex triplet (Math11 Theorem 5); (iii) mass protection from residual valley symmetry (Math12 Lemma 7 and Theorem 8); (iv) local Dirac block reduction $\mathcal{H}_D \cong \mathbb{C}^2_{\mathrm{valley}} \otimes \mathbb{C}^2_{\mathrm{int}}$ (Math13 Proposition 9); (v) reduced Weyl/Dirac witness theorems closing the Class II seed space (Math14 Theorems 11–13 and Corollary 14). The HRR family-count specialisation $\mathrm{ind}(D_E^c) = r - \mu$ at $r = 16$, $\mu = 0$ (Math171-AddA Theorem 15) provides the rank-16 Standard-Model fermion identification at the historical $\mathbb{CP}^2$-route algebraic level.

b. What the present paper does NOT establish. The paper does NOT establish (a) a closed theorem of the form "the chiral family count equals a winding number" on the 3D Brillouin torus — this would require a Callias / Niemi–Semenoff / domain-wall framework on a non-compact extension or an explicit band-projector with controlled IR boundary, which the present archive does not provide; (b) full unconditional Pillar-5 closure — the canonical tier is a candidate mechanism with explicitly stated layered hypotheses; (c) the family-count derivation on the canonical $\Sigma_0 = \mathbb{P}^1 \times \mathbb{P}^1$ Pillar-4 base — this requires the Σ_0 spin-c index calculation (separate Todd class) which is not derived here; (d) the Cl(3) Clifford-algebra closure at the Brillouin-zone-boundary Dirac node — this remains conditional on the Math233 spinor pair-kernel programme.

c. Open follow-ups. The principal open programme item is the `Q-2026-05-02-Pillar5-Mechanism-Rewrite` ticket: a theorem-level rewrite within a Callias-type / Niemi–Semenoff / domain-wall framework that promotes the present mechanism note to an explicit chiral-count theorem with a topological invariant on the appropriate non-compact extension. This is mathematical-research-level work (estimated 2–4 weeks per the `OPEN-QUESTIONS.md` entry) and is the structural mitigation for the previously incorrect $\mathrm{Index}(D) = \mathrm{wind}(m)$ over-claim.

d. Implications for downstream pillars. The Pillar-5 mechanism feeds into Pillar 6 (generations and SM embedding) at the rank-16 specialisation level via Theorem 15 (one SO(10) **16** per chiral irrep). The mass protection from valley symmetry (Theorem 8) supports the candidate Standard-Model neutrino-mass programme at the leading-order level; explicit derivation of neutrino masses from the $\mathbb{C}^2_{\mathrm{int}}$ valley structure is an open programme item. The Pillar-7 quantum-consistency programme (Ward identities, SO(10) anomaly cancellation) is supported by the family-count specialisation but does not in itself force the family count to exactly three; the rank-16 → three-generation question remains the subject of the companion Paper 07 (which is itself currently `[EXTERNAL_USE_FORBIDDEN]` pending the Q-2026-05-02-Pillar7-Three-Generation-Rewrite follow-up).

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